

Examples Summary

Ex 7.1: Summary Statistics of Weight Loss (Mean and Standard Deviation)

The summary statistics indicate that Diet A has a higher average weight loss (mean = 5.341) compared to Diet B (mean = 3.710), with a mean difference of 1.631. This suggests that, on average, individuals following Diet A achieved greater weight loss outcomes than those on Diet B.

The standard deviations reflect the variability within each group. Diet B exhibits slightly higher variability (SD = 2.769) compared to Diet A (SD = 2.536), indicating more dispersed individual outcomes around the mean. This implies that Diet A not only yields greater average weight loss but may also provide more predictable results compared to the more variable outcomes seen with Diet B.

Ex 7.2: Distribution and Quartile Analysis of Weight Loss

These findings offer deeper insights into the distribution of weight loss for both diets. Diet A displays a higher median weight loss (5.642) compared to Diet B (3.745), supporting the earlier observation of a higher mean (5.341 vs. 3.710). This consistency between the mean and median reinforces the idea that Diet A generally leads to better weight loss outcomes.

An examination of the quartiles reveals:

- Diet A's middle 50% of participants recorded weight loss ranging from 3.748 (Q1) to 7.033 (Q3).
- Diet B's middle 50% of participants had weight loss between 1.953 (Q1) and 5.404 (Q3).

The interquartile range (IQR) is relatively similar for both diets—3.285 for Diet A and 3.451 for Diet B—indicating a comparable spread in the central part of the data. However, the slightly larger IQR for Diet B aligns with previous findings of higher variability in its outcomes.

Diet A appears more effective for weight loss based on the following observations:

1. A higher median weight loss (5.642 vs. 3.745).
2. The 25th percentile (Q1) of Diet A (3.748) exceeds the median of Diet B (3.745).
3. The 75th percentile (Q3) of Diet A (7.033) is also higher than that of Diet B (5.404).

These results imply that across all levels of participant response—low, medium, and high—Diet A consistently leads to greater weight loss. Its slightly narrower IQR also indicates more consistent outcomes across participants. Based on these descriptive statistics, Diet A emerges as the more effective and reliable weight-loss intervention compared to Diet B.

Ex 7.3: Consumer Brand Preference by Area

In Area 1, the majority of consumers (60.0%) prefer "Other" cereal brands, while Brand B holds a 24.3% share and Brand A captures the smallest portion at 15.7%. This pattern suggests a strong preference for alternative brands in this demographic, accompanied by weaker loyalty toward the two leading brands.

In contrast, Area 2 presents a more balanced distribution of brand preferences. Although "Other" brands still lead at 45.6%, their dominance is notably reduced compared to Area 1. Together, Brands A and B make up a combined 54.4% of the preferences in Area 2, with Brand B particularly prominent at 33.3%, followed by Brand A at 21.1%.

The most significant differences between the two areas include:

- Brand B's share is higher in Area 2 (33.3%) than in Area 1 (24.3%), representing a 9 percentage point increase.
- Brand A also performs better in Area 2 (21.1%) than in Area 1 (15.7%).
- The market share of "Other" brands is considerably lower in Area 2 (45.6%) than in Area 1 (60.0%).

These differences highlight that Area 2 has a stronger presence and acceptance of major cereal brands, especially Brand B, while Area 1 consumers show a greater inclination toward alternative options.